

Alistair Kung

Software Engineer

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Profile

Experienced software engineer with a background in backend platforms, distributed systems, production web systems, and software quality. Developing deeper specialisation in AI/ML through postgraduate study and practical AI engineering, with particular interest in evaluating where probabilistic systems complement deterministic software and testing practices.

Experience

Currencycloud Ltd/Visa Ltd (Post Acquisition)

London, UK

Software Engineer (Visa Ltd. Aligned Role: Senior Software Engineer)

May 2023 – Sept 2025

- Worked on a high-volume, mission-critical beneficiaries service, with emphasis on resilience, throughput, observability, and maintaining 99.99% availability SLAs.
- Designed and evolved automated testing strategies for production financial systems, applying test-pyramid principles across unit, integration, end-to-end, contract, and mutation testing to balance release confidence, feedback speed, and maintainability.
- Scoped and organised an ongoing rewrite of a tier-0 beneficiaries service from Ruby to Java Spring Boot, using the strangler pattern and Istio Layer 7 routing to maintain availability during migration.
- Led enhanced beneficiary data validations in Java and Spring Boot, lowering manual information requests from downstream payment providers by 80% while supporting payment compliance.
- Used Kafka and the outbox pattern to achieve reliable, idempotent delivery to external providers; also designed a Kafka and Debezium archiving system for the platform's largest database.
- Used TDD and CI/CD quality gates to keep changes testable and releasable, including mutation testing to assess test effectiveness beyond coverage alone.

Full Stack Software Engineer

November 2019 – April 2022

- Built a scalable Anti-Money Laundering system using RabbitMQ, Elasticsearch, Ruby on Rails, and Vue.js.
- Designed an API-first customer onboarding platform adhering to JSON:API standards, with a white-labelled frontend built in React and Next.js.
- Introduced consumer-driven contract testing with Pact across service boundaries and integrated it with Jenkins. This enabled independent API-compatibility checks with less reliance on slower end-to-end environments.

Workshare Ltd

London, UK

Junior Software Engineer

November 2017 – November 2019

- Developed Ruby on Rails backend features and React components for Workshare Transact, a secure web-based file-sharing system for lawyers.
- Resolved high-priority production issues through root-cause analysis and regression-prevention changes in time-sensitive legal workflows.

KPMG, UK

London, UK

Financial Services Audit Associate

October 2014 – January 2017

- Analysed financial controls and risk assessments across banking and insurance clients.

Skills

Languages: Java, Python, Ruby, JavaScript, TypeScript, SQL

Backend & Systems: Spring Boot, Ruby on Rails, Kafka, Debezium, Elasticsearch, RabbitMQ; distributed and event-driven systems

Testing & Quality: TDD, test-pyramid design, unit, integration and end-to-end testing, consumer-driven contract testing (Pact), mutation testing, CI/CD quality gates

AI/ML Engineering: Python, structured LLM outputs, deterministic validation, evaluation design, precision and recall measurement

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Kubernetes, Istio, Jenkins, Git, Datadog

Web: React, Next.js, Vue.js; API-first application design

Selected Projects

Private Client Graph

LLM information extraction and evaluation system

- Built an LLM-based pipeline to extract family and trust relationships from professional documents. Recorded explicit evidence provenance for each relationship and kept probabilistic extraction separate from deterministic validation, normalisation, and graph construction.
 - Designed an evaluation harness comparing generated outputs with hidden ground truth using precision, recall, and provenance accuracy, with deterministic tests around non-probabilistic system behaviour.
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Education

The Chinese University of Hong Kong

MSc Artificial Intelligence

Hong Kong

Sep 2026 – Present

- Postgraduate study across artificial intelligence, machine learning, NLP, generative AI, and agentic systems.

Imperial College London

London, UK

Professional Certificate in Machine Learning & Artificial Intelligence

Dec 2025 – Jun 2026

- Completed practical Python/ML study covering supervised and unsupervised learning, model evaluation, decision trees, support vector machines, neural networks, reinforcement learning, probability, and statistics.

Additional Technical Coursework - Johns Hopkins University

Coursera, Online

Linear Algebra; Probability & Random Variables; Advanced Probability & Statistical Methods

Aug 2025 – Dec 2025

- Coursework in mathematical foundations for machine learning, including matrix methods, random variables, statistical inference, regression, and stochastic processes.

Makers Academy, UK

London, UK

Full-time Software Development Bootcamp

June 2017 – Sept 2017

- Focused on Ruby, JavaScript, web development, TDD, and agile methods; built a Ruby-based single-layer perceptron and training pipeline.

King's College London

London, UK

BSc Business Management (Upper Second-Class Honours)

September 2011 – May 2014